

Kevin Nam (남기빈)

Seoul, South Korea | rallyk3vin@gmail.com | +82 10 2443 7250 | rallykevin.github.io
github.com/SNUSOR-PECT

Last Updated : September 09, 2025

Contents

About Me	1
Research Interests	1
Education	1
Research Experience	1
Major Publications (w. the Applicant's Role)	2
Other Publications	2
Research Projects	3
Teaching Experience	4
Teaching Competencies	4
Honors & Awards	4
Services & Activities	4
Invited Talks	4

About Me

I am a Ph.D. candidate in the Security Optimization Research (SOR) Lab at Seoul National University. My research trajectory began with hardware security topics, including monitoring mechanisms and Trusted Execution Environments (TEE), and has since expanded to privacy-enhancing technologies (PETs). In particular, my most recent works focused on enabling efficient and accurate Machine Learning as a Service over Fully Homomorphic Encryption (FHE).

My research vision is to ***design efficient Security and Privacy Enhancing Computing Systems***. While PETs have achieved significant theoretical and mathematical progress, they are still far from widespread deployment due to their computational complexity. I draw inspiration from the history of Artificial Intelligence: envisioned since the 1970s, AI only became practically deployable with 21st-century hardware and software innovations such as GPUs, distributed systems, and large-scale frameworks. I believe PETs can follow a similar path, becoming practical through retrofitted HW/SW innovations, which is the central goal of my research.

To this end, I pursue what I call non-mathematical approaches—hardware acceleration, compiler optimization, and algorithm–system co-design—to mitigate the performance bottlenecks of PETs, including Homomorphic Encryption, Secure Multi-Party Computation, and TEEs. Ultimately, my goal is to make privacy-preserving computation as conventional and accessible as everyday computing.

Research Interests

- Applied Cryptography
- Systems (HW/SW) for PETs & Cryptography
- HW/SW Security (including confidential computing)
- Security/Privacy for AI Services
- Compiler
- Computer architecture

Education

Seoul National University, MS/Ph.D in Electrical and Computer Engineering Mar 2020 – Feb 2026 (expected)

Advisor : Prof. Yunheung Paek

Dissertation Title (tentative) : Non-Mathematical Optimizations for Fast and Precise Privacy-Preserving Machine Learning System over Encrypted Data

Seoul National University, BS in Electrical and Computer Engineering Mar 2014 – Feb 2020

Left for military service (ROKAF) during 2017-2019

Research Experience

Seoul National University, Seoul, Korea Mar 2020 – Feb 2026*

Research Assistant

Advisor: Yunheung Paek

Yale University, New Haven, CT, USA Jan 2023 – Feb 2023

Visiting Student

Advisor: Jakub Szefer

Post-Quantum-Cryptography Accelerator Design, Side-Channel Attacks on FPGA

Seoul National University, Seoul, Korea Jun 2018 – Dec 2018

Intern

Advisor: Yunheung Paek

HW Implementation of LSTM based Anomalous Branch Behavior Monitoring

Major Publications (w. the Applicant's Role)

- [1] **[Security'25]** "SLOTHE: Lazy Approximation of Non-Arithmetic Neural Network Functions over Encrypted Data". Kevin Nam*, Youyeon Joo*, Seungjin Ha, and Yunheung Paek. (* co-first authors) In *USENIX Security Symposium (Security'25)*, August 2025. (KIISE S, BK21 IF: 3, CSRankings).
Applicant's role. As a first author, I developed the core idea of 'lazy approximation' based on code analysis. My primary responsibility was to design and implement the compiler-based framework that analyzes and transforms the neural network code. I also led the experimental evaluation, and wrote almost all the manuscript and coordinated co-author contributions.
- [2] **[ASPLOS'25]** "Affinity-based Optimizations for TFHE on Processing-in-DRAM". Kevin Nam, Heonhui Jung, Hyunyoung Oh, and Yunheung Paek. In the *30th International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS'25)*, April 2025. (KIISE S, BK21 IF: 4, CSRankings).
Applicant's role. As the first author, I suggested the key insight of TFHE algorithm translation which makes TFHE fit for Processing-in-memory systems, conceived and developed the central idea of using data affinity to optimize TFHE on Processing-in-DRAM systems. I was responsible for the entire research pipeline, including designing the full HW/SW stacks, leading all performance benchmarks and paper writing.
- [3] **[Security'25]** "LOHEN: Layer-wise Optimizations for Neural Network Inferences over Encrypted Data with High Performance or Accuracy". Kevin Nam, Youyeon Joo, Dongju Lee, Seungjin Ha, Hyunyoung Oh, Hyungon Moon, and Yunheung Paek. In *USENIX Security Symposium (Security'25)*, August 2025. (KIISE S, BK21 IF: 3, CSRankings).
Applicant's role. As the first author, I proposed the problem formulation, developed the cryptographic gadgets and optimization methodology. I implemented the experimental framework with the co-authors. I led the evaluation on large-scale neural networks. I wrote the main body of the paper with the guidance of two professors.
- [4] **[ICCAD'22]** "Accelerating N-bit Operations over TFHE on Commodity CPU-FPGA". Kevin Nam, Hyunyoung Oh, Hyungon Moon, and Yunheung Paek. In *IEEE/ACM International Conference on Computer-Aided Design (ICCAD'22)*, November 2022. (KIISE A, BK21 IF: 3, CSRankings).
Applicant's role. As the first author, I designed the hardware architecture and parallel dataflow for accelerating TFHE operations on a commodity CPU-FPGA platform. I performed the end-to-end comprehensive evaluation and wrote the main body of the manuscript with the guidance of two professors.
- [5] "MeetGo: A trusted execution environment for remote applications on FPGA". Hyunyoung Oh, Kevin Nam, Seongil Jeon, and Yeongpil Cho, Yunheung Paek. In *IEEE Access*, 2021.
Applicant's role. As a co-author, I contributed to the FPGA implementation process of the design and ran experiments (neural network and crypto-wallet). I also participated in writing parts of the design and evaluation sections, and refining the paper at the final stage.

Other Publications

Refereed Conference Publications

- [6] "Area-Efficient Accelerator for the Full NTRU-KEM Algorithm". Yongseok Lee, Kevin Nam, Youyeon Joo, Jeehwan Kim, Hyunyoung Oh, Yunheung Paek. In *International Conference on Computational Science and Its Applications (ICCSA'23)*. Cham: Springer Nature Switzerland, 2023.

Refereed Journal Publications

- [7] "An Efficient Hardware/Software Co-design for FALCON on Low-End Embedded Systems". Yongseok Lee, Jonghee Youn, Kevin Nam, Heon Hui Jung, Myunghyun Cho, Jimyung Na, Jong-Yeon Park, Seungsu Jeon, Bo Gyeong Kang, Hyunyoung Oh, Yunheung Paek. In *IEEE Access*, 2024.

Refereed Poster Publications

- [8] "Affinity-based Optimizations of Homomorphic Encryption Operations on Processing-in-DRAM". Kevin Nam, Heonhui Jung, Hyunyoung Oh, and Yunheung Paek. In the *61th ACM/IEEE Design Automation Conference Work-in-Progress (DAC'24 WiP)*, 2024.
- [9] "Implementing Efficient, Precise N-bit Operations of TFHE on Commodity CPU-FPGA". Kevin Nam, Youyeon Joo, Dongju Lee, Seungjin Ha, Hyunyoung Oh, Hyungon Moon, and Yunheung Paek. In the *59th ACM/IEEE Design Automation Conference Work-in-Progress (DAC'22 WiP)*, 2022.

Domestic Papers (first author ones only)

- [10] "Quantization of Non-Arithmetics to Optimize CNNs over TFHE". Kevin Nam, Heonhui Jung, Dongju Lee, Yunheung Paek. Annual Symposium of KIPS, 2024
- [11] "Intermediate Data Guided Approximation for Fast and Accurate Encrypted Neural Networks". Kevin Nam, Youyeon Joo, Seungjin Ha, Yunheung Paek. Annual Conference of KIPS, 2024.
- [12] "Side-Channel Attacks on Homomorphic Encryption and Their Mitigation Methods". Kevin Nam, Youyeon Joo, Yunheung Paek. Annual Symposium of KIPS, 2023 (**NIPA Director Award**)
- [13] "Realization of Homomorphic Encrypted Deep Learning Models". Kevin Nam, Myunghyun Cho, Hyunjun Kim, Yunheung Paek. Annual Conference of KIPS, 2021, (**Undang Academic Award**)
- [14] "Partially Homomorphic Encryption HW accelerator". Kevin Nam, Jiwon Chang, Myunghyun Cho, Inyoung Bang, Yunheung Paek. Annual Conference of KIPS, 2020

Patents

- [15] "METHOD FOR PROCESSING HOMOMORPHIC CIPHERTEXT AND ELECTRONIC APPARATUS". Yunheung Paek, Heonhui Jung, Kevin Nam, Seungjin Ha, Junbum Shin, Inkwan Yu, Sunchul Jung, Jungjoo Seo. Feb 2025. US Patent No. 19/052442, Applied
- [16] "METHOD FOR PROCESSING HOMOMORPHIC CIPHERTEXT AND ELECTRONIC APPARATUS". Yunheung Paek, Heonhui Jung, Kevin Nam, Seungjin Ha, Junbum Shin, Inkwan Yu, Sunchul Jung, Jungjoo Seo. Jan 2025. KR Patent No. 10-2025-0008424, Applied
- [17] "METHOD AND APPARATUS FOR COMPUTING ENCRYPTED DATA USING MULTI-HOMOMORPHIC ENCRYPTION SYSTEM". Yunheung Paek, Kevin Nam, Youyeon Joo. Nov 2024. KR Patent No. 10-2024-0153707, Applied
- [18] "TEE ENVIRONMENT PROVIDING APPARATUS AND METHOD USING FPGA". Yunheung Paek, Hyunyoung Oh, Kevin Nam, Yeongpil Cho. Jan 2021. KR Patent No. 10-2021-0006281, Applied

Research Projects

w. fundings, Principal Investigator : Yunheung Paek.

Privacy-Preserving Neural Network Inference System funded by NRF, South Korea	Oct 2023 – current
• Integrate non-mathematical methods (e.g., compiler) to enhance the performance of cryptographic backends	
Secure FHE Key Management System funded by CryptoLab, South Korea	Mar 2023 – April 2024
• Designed efficient TEE-based FHE key management system within SGX restrictions	
Neural Network Program to FHE Transpiler funded by ETRI, South Korea	Oct 2023 – Nov 2023
• Designed a transpiler that translates tensorflow/Pytorch programs into FHE programs	
PEC Platform funded by the National Intelligence Service, South Korea	Mar 2022 – Nov 2023
• Developed a framework that analyzes normal codes and suggest appropriate PETs and generates a secure code	
FPGA Accelerator for CKKS funded by CryptoLab, South Korea	Mar 2022 – Nov 2022
• Participated as the algorithm analyzer - designed the overall operation mapping and pipeline for CKKS on FPGA	
FPGA Accelerator for TFHE funded by the National Intelligence Service	Mar 2021 – Nov 2021
• Designed an FPGA-based TFHE accelerator exploiting the double parallelism of TFHE	
Designing HW/SW-based Computing Base for Secure Boxes funded by IITP, South Korea	Sep 2021 – current
• Implemented HW/SW-based isolated VMs with TEEs and HE	
Designing PQC Accelerators funded by Samsung Electronics, South Korea	Sep 2020 – Sep 2024
• Integrated and optimized the HW modules	
FPGA-based Trusted Execution Environments funded by NRF, South Korea	Mar 2020 – July 2022
• Implemented TEE isolation systems on FPGAs	

Teaching Experience

Seoul National University

- (Teaching Assistant) Logic Circuit Design Course, 430.201A
Mar 2020 - Aug 2020, Sep 2022 - Dec 2022, Sep 2023 - Dec 2023, Sep 2024 - Dec 2024
- (Teaching Assistant) Data Security & Privacy, 430.658
Sep 2024 - Dec 2024, Sep 2025 - Dec 2025
- (Teaching Assistant) Creative Design Project, 430.405
Mar 2022 - current

Teaching Competencies

I am prepared and enthusiastic to teach a broad range of courses across computer science and engineering, from foundational undergraduate subjects to specialized graduate seminars.

Computer Systems & Architecture

- Computer Architecture
- Compiler
- Parallel Computing
- Customized Computer Design
- Logic/Digital Circuit Design

Core CS & Programming

- System Programming
- Data Structure / Algorithm

AI Related Courses

- Intro. to ML
- AI Acc. Design
- Intro to AI security

Security & Privacy

- Computer Security
- Applied/Modern Cryptography
- Systems for AI Security & Privacy

Graduate / Seminar Courses

- Data Security & Privacy
- Advanced Seminar on PETs
- Graduate Seminar on AI Security

Furthermore, I am a strong proponent of *project-based learning* and am keen to guide students in design projects with topics drawn from my research in systems and security.

Honors & Awards

Excellence Paper Award , from ACK 2024	May 2024
Award for Excellence in Teaching Assistant , from Seoul National University <i>Logic circuit Design Course</i>	Jan 2024
NIPA Director Award from ASK 2023	Jun 2023
Undang Academic Award from KIPS, <i>Best Graduate Student Paper Award</i>	Dec 2021
Best Paper Award from ASK 2021	May 2021
Scholarship BK 21+ Scholarship by the Ministry of Education of Korea	Sep 2020 - Feb 2021 Mar 2023 - present

Services & Activities

IEEE Symposium on Security and Privacy 2026 Artifact Evaluation Committee

EuroSys 2025 Shadow Program Committee

CES 2017 Exhibitor (BOM Team)

Invited Talks

- [1] "Zero-Error Privacy-Preserving Machine Learning as a Service" presented at Seoul National University AI Health & Care Center Symposium, December 2023
- [2] "Fully Homomorphic Encryption based Neural Network Inference" presented at Ewha Women University, September 2022